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Requirements for this paper/Benodighede vir hierdie vraestel:				Resources/Hulpmiddels:
Answer Scripts/ Antwoordskrifte:		Multi-choice cards (A4)/ Multikeusekaarte (A4)		
Attendance Slips (Fill-in Paper)/ Presensiestrokies (Invulvraestel):		Graph Paper/ GrafiekPapier		
Scrap Paper/ Rofwerkpapier		Calculators/ Sakrekenaars		
Multi-choice cards (A5)/ Multikeusekaarte (A5)		Laptop (Power not provided)/ Skootrekenaar (Krag word nie voorsien nie)		

Type of Assessment/
Tipe Assessering:

Duration/
Tydsduur:

Paper Number/
Vraestel Nommer:

Maximum Marks/
Maksimum Punte:

Module Code/
Modulekode:

Module Description/
Module Beskrywing:

Examiner(s)/
Eksaminator(e):

Date/
Datum:

Internal/Interne
Moderator(s):

Time/
Tyd:

External Moderator(s)/
Eksterne Moderator(s):

Qualification/
Kwalifikasie:

Submission of answer scripts/Inhandiging van antwoordskrifte:

Instructions/Instruksies

2023 CMPG311 1st Opportunity - ENGLISH

Question 1 (20)

1.1. Name and briefly describe four consequences of data redundancy. [8]

1.2. Study the table below. What data redundancies do you detect? How could these redundancies lead to anomalies? [6]

PAT_NUM	PAT_LNAME	PAT_FNAME	DOC_LNAME	DRUG_NAME
100	Kolmycz	George	Dreifag	Afgapan-15
101	Lewis	Rhonda	Chin	Droalene Chloride
102	Vandam	Rhet	Dreifag	Afgapan-15
103	Jones	Anna	Chin	Droalene Chloride
104	Lnage	John	Minh	Afgapan-25
105	Williams	Robert	Dreifag	Droalene Chloride

1.3. Draw a diagram to indicate the main steps (phases) of the database life cycle (DBLC). [6]

Question 2 (30)

A martial arts school needs a database to store their information.

Create a complete Crow's Foot ERD for these requirements:

- Students are given a student number when they join the school. This is stored along with their name, date of birth, and the date they joined the school.
- All instructors are also students, but clearly, not all students are instructors. In addition to the normal student information, for each instructor, the date that they start working as an instructor must be recorded, along with their instructor status (compensated or volunteer).
- An instructor may be assigned to teach any number of classes, but each class has one and only one assigned instructor. Some instructors, especially volunteer instructors, may not be assigned to any class.
- A student will attend many different class meetings; and each class meeting is normally attended by many students. Some class meetings may have no students show up for that meeting. New students may not have attended any class meetings yet.
- At any given meeting of a class, instructors other than the assigned instructor may show up to help. Therefore, a given class meeting may have several instructors (a head instructor and many assistant instructors), but it will always have at least the one instructor that is assigned to that class.
- Each student holds a rank in the martial arts. The rank name, belt colour, and rank requirements are stored. Each rank will have numerous rank requirements. Each requirement is considered a requirement just for the rank at which the requirement is introduced.
- A given rank may be held by many students. While it is customary to think of a student as having a single rank, it is necessary to track each student's progress through the ranks. Therefore, every rank that a student attains is kept in the system. New students joining the school are automatically given a white belt rank. The date that a student is awarded each rank should be kept in the system. All ranks have at least one student that has achieved that rank at some time.

Question 3

(20)

PAT_NUM	DOC_NUM	PAT_LNAME	PAT_FNAME	DOC_LNAME	DRUG_NUM	DRUG_NAME	DOSAGE
100	001	Kolmycz	George	Dreifag	005	Afgapan-15	1/day
101	002	Lewis	Rhonda	Chin	007	Droalene Chloride	2/day
102	001	Vandam	Rhet	Dreifag	005	Afgapan-15	1/day
103	002	Jones	Anna	Chin	003	Droalene Chloride	1/day
104	003	Lnage	John	Minh	007	Afgapan-25	3/day
105	001	Williams	Robert	Dreifag	003	Droalene Chloride	2/day

3.1 Write the relational schema, draw its dependency diagram and identify all dependencies, including all partial and transitive dependencies. You can assume that the table does not contain repeating groups. (Hint: This table uses a composite primary key.)

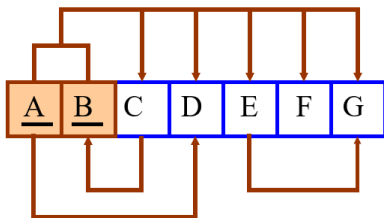
(6)

3.2 Normalise the dependency diagram you created above (in 3.1) to 3NF.

(6)

3.3 Use the provide dependency diagram and produce a collection of dependency diagrams that are all in 3NF.

(8)



Question 4

(30)

Use the provided tables to write SQL queries to address the following requests.

Table 1 - **ASSIGNMENT**

ASSIGN_NUM	ASSIGN_DATE	PROJ_NUM	EMP_NUM	ASSIGN_JOB	ASSIGN_CHG_HR	ASSIGN_HOURS	ASSIGN_CHARGE
1001	2018/03/22	18	103	503	84,5	3,5	295,75
1002	2018/03/22	22	117	509	34,55	4,2	145,11
1003	2018/03/22	18	117	509	34,55	2	69,1
1004	2018/03/22	18	103	503	84,5	5,9	498,55
1005	2018/03/22	25	108	501	96,75	2,2	212,85
1006	2018/03/22	22	104	501	96,75	4,2	406,35
1007	2018/03/22	25	113	508	50,75	3,8	192,85
1008	2018/03/22	18	103	503	84,5	0,9	76,05
1009	2018/03/23	15	115	501	96,75	5,6	541,8
1010	2018/03/23	15	117	509	34,55	2,4	82,92
1011	2018/03/23	25	105	502	105	4,3	451,5
1012	2018/03/23	18	108	501	96,75	3,4	328,95
1013	2018/03/23	25	115	501	96,75	2	193,5
1014	2018/03/23	22	104	501	96,75	2,8	270,9
1015	2018/03/23	15	103	503	84,5	6,1	515,45
1016	2018/03/23	22	105	502	105	4,7	493,5
1017	2018/03/23	18	117	509	34,55	3,8	131,29
1018	2018/03/23	25	117	509	34,55	2,2	76,01
1019	2018/03/24	25	104	501	110,5	4,9	541,45
1020	2018/03/24	15	101	502	125	3,1	387,5
1021	2018/03/24	22	108	501	110,5	2,7	298,35
1022	2018/03/24	22	115	501	110,5	4,9	541,45
1023	2018/03/24	22	105	502	125	3,5	437,5
1024	2018/03/24	15	103	503	84,5	3,3	278,85
1025	2018/03/24	18	117	509	34,55	4,2	145,11

Table 2 - **EMPLOYEE**

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIREDATE	JOB_CODE	EMP_YEARS
101	News	John	G	2000/11/08	502	14
102	Senior	David	H	1989/07/12	501	25
103	Arbough	June	E	1996/12/01	500	18
104	Ramoras	Anne	K	1987/11/15	501	27
105	Johnson	Alice	K	1993/02/01	502	22
106	Smithfield	William		2004/06/22	500	10
107	Alonzo	Maria	D	1993/10/10	500	21
108	Washington	Ralph	B	1991/08/22	501	23
109	Smith	Larry	W	1997/07/18	501	17
110	Olenko	Gerald	A	1995/12/11	505	19
111	Wabash	Geoff	B	1991/04/04	506	24
112	Smithson	Darlene	M	1994/10/23	507	20
113	Joebrood	Delbert	K	1996/11/15	508	18
114	Jones	Annelise		1993/08/20	508	21
115	Bawangi	Travis	B	1992/01/25	501	23
116	Pratt	Gerald	L	1997/03/05	510	18
117	Williamson	Angie	H	1996/06/19	509	18
118	Frommer	James	J	2005/01/04	510	10

Table 3 - **JOB**

JOB_CODE	JOB_DESCRIPTION	JOB_CHG_HOUR	JOB_LAST_UPDATE
500	Programmer	35,75	2017/11/20
501	Systems Analyst	96,75	2017/11/20
502	Database Designer	125	2018/03/24
503	Electrical Engineer	84,5	2017/11/20
504	Mechanical Engineer	67,90000153	2017/11/20
505	Civil Engineer	55,77999878	2017/11/20
506	Clerical Support	26,87000084	2017/11/20
507	DSS Analyst	45,95000076	2017/11/20
508	Applications Designer	48,09999847	2018/03/24
509	Bio Technician	34,54999924	2017/11/20
510	General Support	18,36000061	2017/11/20

Table 4 - **PROJECT**

PROJ_NUM	PROJ_NAME	PROJ_VALUE	PROJ_BALANCE	EMP_NUM
15	Evergreen	1453500	1002350	103
18	Amber Wave	3500500	2110346	108
22	Rolling Tide	805000	500345,2	102
25	Starflight	2650500	2309880	107

- 4.1 Display the projects names of the projects that employee with number 103 have worked on. Duplicate values should not be displayed. (4)
- 4.2 Display the project number, dates, and jobs of all the project where the charge of the project exceeded 400. (3)
- 4.3 Display the job with the lowest charge per hour. (2)
- 4.4 Display the names of all employees that are exactly 4 characters long. (2)
- 4.5 Display the Total charge for project number 18 as "Total Charge". (3)
- 4.6 Display the Total charge for each project according to the project number, sorted by the project number. (5)
- 4.7 Display the job description of the jobs assigned to project no 18. (5)
- 4.8 Display the names of all the project learners for the projects in which employee 117 worked, no duplicated values must be displayed. (6)

TOTAL/TOTAAL: 100

Instructions/ Instruksies

2023 CMPG311 1^{ste} Geleentheid - AFRIKAANS

Vraag 1 (20)

1.1. Noem en beskryf kortliks vier gevolge van data-oortolligheid. [8]

1.2. Bestudeer die tabel hieronder. Watter data-oortolligheid bespeur jy? Hoe kan hierdie afdankings tot onreëlmatighede lei? [6]

PAT_NUM	PAT_LNAME	PAT_FNAME	DOC_LNAME	DRUG_NAME
100	Kolmycz	George	Dreifag	Afgapan-15
101	Lewis	Rhonda	Chin	Droalene Chloride
102	Vandam	Rhet	Dreifag	Afgapan-15
103	Jones	Anna	Chin	Droalene Chloride
104	Lnage	John	Minh	Afgapan-25
105	Williams	Robert	Dreifag	Droalene Chloride

1.3. Teken 'n diagram om die hoofstappe (fases) van die databasislewensiklus (DBLC) aan te dui. [6]

Vraag 2 (30)

'n Gevegkuns- (*Martial arts*) skool benodig 'n databasis om hul inligting te stoor.

Skep 'n volledige Crow's Foot ERD vir hierdie vereistes:

- Studente kry 'n studentenommer wanneer hulle by die skool aansluit. Dit word saam met hul naam, geboortedatum en die datum waarop hulle by die skool aangesluit het gestoor.
- Alle instruktors is ook studente, maar dit is duidelik dat nie alle studente instruktors is nie. Benewens die normale studente-inligting, moet vir elke instrukteur die datum waarop hulle as 'n instrukteur begin werk, aangeteken word, saam met hul instrukteurstatus (vergoed of vrywilliger).
- 'n Instrukteur kan aangewys word om enige aantal klasse te gee, maar elke klas het een en slegs een toegewysde instrukteur. Sommige instruktors, veral vrywillige instruktors, word nie altyd aan 'n klas toegewys nie.
- 'n Student sal baie verskillende klasse bywoon; en elke klas word normaalweg deur baie studente bygewoon. Sommige klasvergaderings mag dalk geen studente vir daardie vergadering opdaag nie. Nuwe studente het dalk nog nie enige klasvergaderings bygewoon nie.
- By enige gegewe klas kan 'n ander instruktors as die aangewese instrukteur opdaag om te help. Daarom kan 'n gegewe klas verskeie instruktors hê ('n hoofinstrukteur en baie assistent-instruktors), maar dit sal altyd ten minste die een instrukteur hê wat aan daardie klas toegewys is.
- Elke student het 'n rang in die gevegskuns. Die rangnaam, bandkleur en rangvereistes word gestoor. Elke rang sal talle rangvereistes hê. Elke vereiste word as 'n vereiste beskou net vir die rang waarop die vereiste ingestel word.
- 'n Gegewe rang kan deur baie studente bekleed word. Alhoewel dit gebruiklik is om daaraan te dink dat 'n student 'n enkele rang het, en is dit nodig om elke student se vordering deur die range na te spoor. Daarom word elke rang wat 'n student behaal in die stelsel gehou. Nuwe studente wat by die skool aansluit, kry outomaties 'n witgordelrang. Die datum waarop 'n student elke rang toegeken word, moet in die stelsel gehou word. Alle range het ten minste een student wat daardie rang een of ander tyd behaal het.

Vraag 3

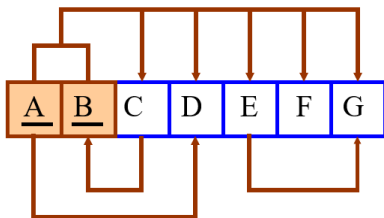
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PAT_NUM	DOC_NUM	PAT_LNAME	PAT_FNAME	DOC_LNAME	DRUG_NUM	DRUG_NAME	DOSAGE
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101	002	Lewis	Rhonda	Chin	007	Droalene Chloride	2/day
102	001	Vandam	Rhet	Dreifag	005	Afgapan-15	1/day
103	002	Jones	Anna	Chin	003	Droalene Chloride	1/day
104	003	Lnage	John	Minh	007	Afgapan-25	3/day
105	001	Williams	Robert	Dreifag	003	Droalene Chloride	2/day

3.1 Skryf die relasionele skema, teken sy afhanklikheidsdiagram en identifiseer alle afhanklikhede, insluitend alle gedeeltelike en oorganklike afhanklikhede. Jy kan aanvaar dat die tabel nie herhalende groepe bevat nie. (Wenk : Hierdie tabel gebruik 'n saamgestelde primêre sleutel.) (6)

3.2 Normaliseer die afhanklikheidsdiagram wat jy hierbo geskep het (in 3.1) na 3NF. (6)

3.3 Gebruik die onderstaande afhanklikheidsdiagram en produseer 'n versameling afhanklikheidsdiagramme wat almal in 3NF is. (8)



Vraag 4

(30)

Gebruik die verskafde tabelle om SQL-navrae te skryf om die volgende versoeke aan te spreek.

Tabel 1- **ASSIGNMENT**

ASSIGN_NUM	ASSIGN_DATE	PROJ_NUM	EMP_NUM	ASSIGN_JOB	ASSIGN_CHG_HR	ASSIGN_HOURS	ASSIGN_CHARGE
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1004	2018/03/22	18	103	503	84,5	5,9	498,55
1005	2018/03/22	25	108	501	96,75	2,2	212,85
1006	2018/03/22	22	104	501	96,75	4,2	406,35
1007	2018/03/22	25	113	508	50,75	3,8	192,85
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1009	2018/03/23	15	115	501	96,75	5,6	541,8
1010	2018/03/23	15	117	509	34,55	2,4	82,92
1011	2018/03/23	25	105	502	105	4,3	451,5
1012	2018/03/23	18	108	501	96,75	3,4	328,95
1013	2018/03/23	25	115	501	96,75	2	193,5
1014	2018/03/23	22	104	501	96,75	2,8	270,9
1015	2018/03/23	15	103	503	84,5	6,1	515,45
1016	2018/03/23	22	105	502	105	4,7	493,5
1017	2018/03/23	18	117	509	34,55	3,8	131,29
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1019	2018/03/24	25	104	501	110,5	4,9	541,45
1020	2018/03/24	15	101	502	125	3,1	387,5
1021	2018/03/24	22	108	501	110,5	2,7	298,35
1022	2018/03/24	22	115	501	110,5	4,9	541,45
1023	2018/03/24	22	105	502	125	3,5	437,5
1024	2018/03/24	15	103	503	84,5	3,3	278,85
1025	2018/03/24	18	117	509	34,55	4,2	145,11

Tafel 2- **EMPLOYEE**

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIREDATE	JOB_CODE	EMP_YEARS
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102	Senior	David	H	1989/07/12	501	25
103	Arbough	June	E	1996/12/01	500	18
104	Ramoras	Anne	K	1987/11/15	501	27
105	Johnson	Alice	K	1993/02/01	502	22
106	Smithfield	William		2004/06/22	500	10
107	Alonzo	Maria	D	1993/10/10	500	21
108	Washington	Ralph	B	1991/08/22	501	23
109	Smith	Larry	W	1997/07/18	501	17
110	Olenko	Gerald	A	1995/12/11	505	19
111	Wabash	Geoff	B	1991/04/04	506	24
112	Smithson	Darlene	M	1994/10/23	507	20
113	Joebrood	Delbert	K	1996/11/15	508	18
114	Jones	Annelise		1993/08/20	508	21
115	Bawangi	Travis	B	1992/01/25	501	23
116	Pratt	Gerald	L	1997/03/05	510	18
117	Williamson	Angie	H	1996/06/19	509	18
118	Frommer	James	J	2005/01/04	510	10

Tafel 3 - **JOB**

JOB_CODE	JOB_DESCRIPTION	JOB_CHG_HOUR	JOB_LAST_UPDATE
500	Programmer	35,75	2017/11/20
501	Systems Analyst	96,75	2017/11/20
502	Database Designer	125	2018/03/24
503	Electrical Engineer	84,5	2017/11/20
504	Mechanical Engineer	67,90000153	2017/11/20
505	Civil Engineer	55,77999878	2017/11/20
506	Clerical Support	26,87000084	2017/11/20
507	DSS Analyst	45,95000076	2017/11/20
508	Applications Designer	48,09999847	2018/03/24
509	Bio Technician	34,54999924	2017/11/20
510	General Support	18,36000061	2017/11/20

Tabel 4- **PROJECT**

PROJ_NUM	PROJ_NAME	PROJ_VALUE	PROJ_BALANCE	EMP_NUM
15	Evergreen	1453500	1002350	103
18	Amber Wave	3500500	2110346	108
22	Rolling Tide	805000	500345,2	102
25	Starflight	2650500	2309880	107

- 4.1 Vertoon die projekte se name van die projekte waaraan werknemer met nommer 103 gewerk het. Duplikaatwaardes moet nie vertoon word nie. (4)
- 4.2 Vertoon die projeknommer, datums en take (jobs) van al die projekte waar die koste van die projek 400 oorskry het. (3)
- 4.3 Vertoon die werk met die laagste heffing (cost) per uur. (2)
- 4.4 Vertoon die name van alle werknemers wat presies 4 karakters lank is. (2)
- 4.5 Vertoon die Totale heffing vir projek nommer 18 as "Totale heffing". (3)
- 4.6 Vertoon die Totale heffing vir elke projek volgens die projeknommer, wat gesorteer is volgens die projeknommer. (5)
- 4.7 Vertoon die posbeskrywing van die poste wat aan projek nr. 18 toegewys is. (5)
- 4.8 Vertoon die name van al die projekteleiders vir die projekte waarin werknemer 117 gewerk het, geen gedupliseerde waardes moet vertoon word nie. (6)

TOTAL/TOTAAL: 100